Test 6 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature

1. Reading Comprehension

Read the passage and answer the questions:

"During the school science exhibition, Farhan presented a project on water purification using simple materials. Visitors appreciated how the model demonstrated filtration and conservation. The judges praised his clarity and confidence while explaining the process."

a) What was the theme of Farhan's project?
b) How did visitors react to the presentation?
c) Which personal qualities were noticed by the judges?
d) Write a suitable title for the passage.

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2. Grammar and Usage

a) Convert into indirect speech:
The teacher said, "Submit your assignment tomorrow."

b) Identify the tense:
"They had completed the task before sunset."

c) Fill in the blank with a suitable preposition:
She succeeded ___ hard work and patience.

d) Rewrite using comparative degree:
"This road is the safest route in the city."

e) Write a sentence using the word *innovation*.

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3. Writing Skills

Write an essay of 150–180 words on:
"The Role of Technology in Improving Education"

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4. Literature Analysis

a) Explain how dialogue helps reveal character traits.
b) Describe one external conflict commonly found in novels.
c) What is symbolism? Give one original example.

Section B: Mathematics

1. Algebra

a) Solve for x:
$7x - 4 = 3x + 20$

b) Factorize fully:
$2y^2 - 11y - 6$

c) Simplify:
$(3a^2b)(4ab^3)$

d) If $5m - 2n = 16$ and $m - 2n = 4$, find m and n.

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2. Geometry

a) Find the volume of a cuboid with dimensions 8 cm, 5 cm, and 4 cm.
b) Calculate the interior angle of a regular hexagon.
c) Determine the distance between points (−3, 2) and (5, −4).

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3. Trigonometry

a) Evaluate: cos 45° − sin 45°
b) If sin θ = 3/5, find tan θ.
c) A ladder leans against a wall making an angle of 60° with the ground. If its length is 10 m, find the height reached on the wall.

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4. Commercial Mathematics

a) Find the profit percent if a bicycle costs $180 and sells for $225.
b) Divide $720 in the ratio 3:5.
c) Find the compound interest on $1000 at 10% for 2 years.

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5. Statistics and Probability

a) Find the mean of: 6, 9, 11, 14, 20
b) Draw a bar graph for the following data:
Books read in a month: Ali 4, Sara 7, Omar 5
c) A die is thrown once. Find the probability of getting a multiple of 3.

Section C: Science

1. Biology

a) Describe the function of red blood cells.
b) Differentiate between autotrophic and heterotrophic nutrition.
c) Explain the process of respiration in humans.
d) Define adaptation with one animal example.
e) What role do producers play in a food chain?

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2. Physics

a) Define work and state its SI unit.
b) A car accelerates from rest to 20 m/s in 5 seconds. Find acceleration.
c) Why does a coin fall when a card beneath it is pulled quickly?
d) Explain conduction of heat.
e) State one application of refraction in daily life.

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3. Chemistry

a) Distinguish between elements and compounds.
b) Write the chemical formula of washing soda.
c) Define pH scale.
d) State two uses of acids in industry.
e) What happens during neutralization?

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4. Environmental Science

a) Define renewable resources.
b) Explain causes of air pollution.
c) What is waste management?
d) Suggest two ways to conserve water at home.

Section D: Social Studies

1. History

a) What was the Industrial Revolution?
b) State two impacts of colonialism on Asia.
c) Explain the significance of the United Nations.

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2. Geography

a) Define plate tectonics.
b) Name one desert of Africa and describe its climate.
c) How do mountains influence rainfall?

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3. Civics / Political Science

a) What is universal adult franchise?
b) Explain the importance of freedom of expression.
c) State one responsibility of a citizen.

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4. Economics

a) Define inflation.
b) What is unemployment?
c) Explain the concept of supply with one example.

Section E: ICT / Computer Studies

1. Computer Systems

a) What is hardware? Give two examples.
b) Explain the role of CPU.

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2. Software

a) What is presentation software used for?
b) Name one advantage of cloud storage.

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3. Programming Logic

a) Define a flowchart.
b) Write a condition to check if a number is positive.

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4. Cyber Awareness

a) What is phishing?
b) State one way to create a strong password.

Section F: Islamiyat

1. Quranic Studies

a) State the central message of Surah Al-Asr.
b) Mention one Quranic instruction about honesty.

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2. Beliefs

a) Explain belief in the Day of Judgment.
b) Why is Tawheed fundamental in Islam?

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3. Worship

a) What is fasting and its purpose?
b) Name two conditions for the obligation of Hajj.

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4. Islamic Ethics

a) How does Islam discourage dishonesty?
b) Explain the concept of brotherhood in Islam.

Section G: Arts and Creativity

1. Art Education

a) Define texture in art.
b) Name one drawing tool and its use.

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2. Cultural Studies

a) What is folk art?
b) Name one traditional craft from any country.

Section H: Health & Physical Education

1. Physical Fitness

a) What is flexibility?
b) Name one exercise that improves balance.

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2. Nutrition

a) Define a balanced diet.
b) Why are proteins important for growth?

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3. Safety and First Aid

a) What should be done for a minor burn?
b) State one rule of playground safety.